CS486/686: Introduction to Artificial Intelligence
Lecture 9b - Planning with Uncertainty (II)

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Readings: Poole & Mackworth Chap. 12.5

Agents as Processes

Agents carry out actions:

- forever (**infinite horizon**)
- until some stopping criteria is met (**indefinite horizon**)
- finite and fixed number of steps (**finite horizon**)

Decision-Theoretic Planning

What should an agent do when

- it gets rewards (and punishments) and tries to **maximize its rewards** received
- actions can be noisy; the **outcome of an action can't be fully predicted**
- there is a model that specifies the **probabilistic outcome of actions**
- the world is **fully observable**: the current state is always fully in evidence

for the various planning horizons?

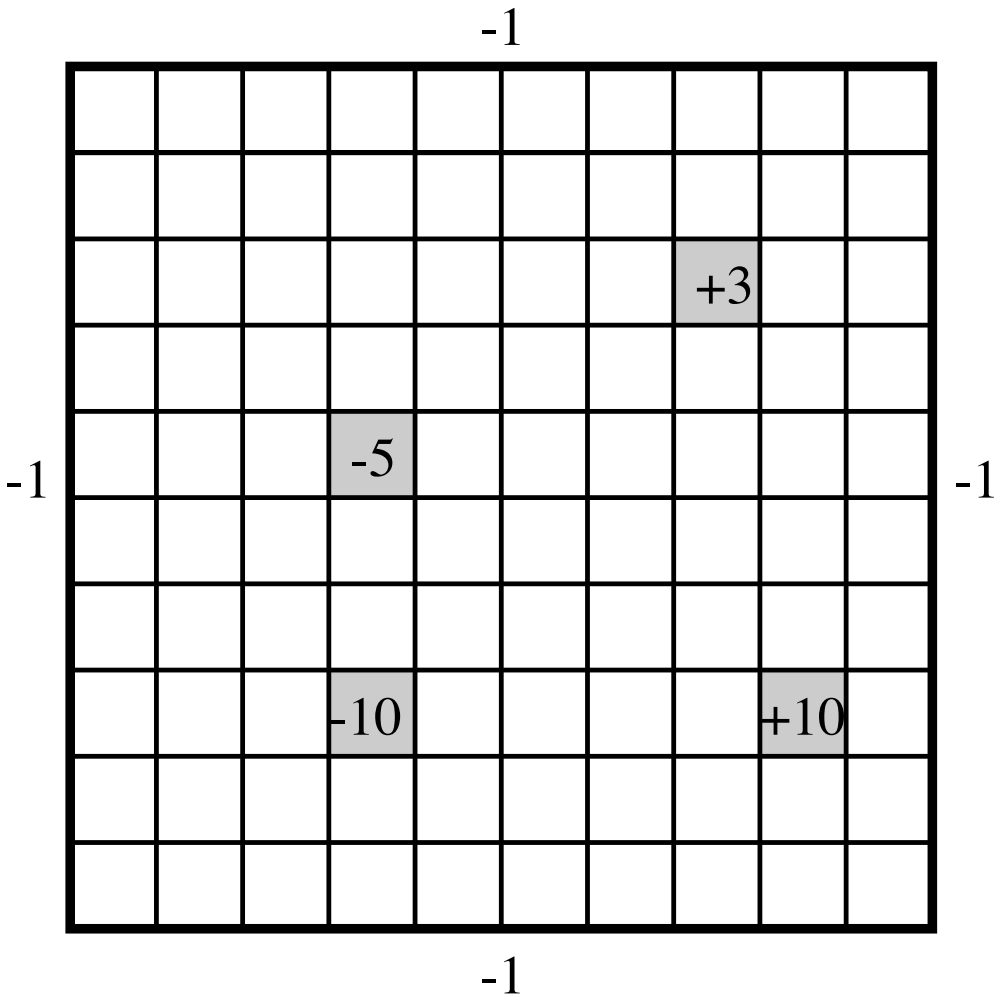
World State

- The world state is the information such that if you knew the world state, no information about the past is relevant to the future. **Markovian assumption**.
- Let S_i, A_i be the state,action at time i

$$P(S_{t+1} \mid S_0, A_0, \ldots, S_t, A_t) = P(S_{t+1} \mid S_t, A_t)$$

  $P(s' \mid s, a)$ is the probability that the agent will be in state s' immediately after doing action a in state s.
- The dynamics is **stationary** if the distribution is the same for each time point.

Example: Simple Grid World



Grid World Model

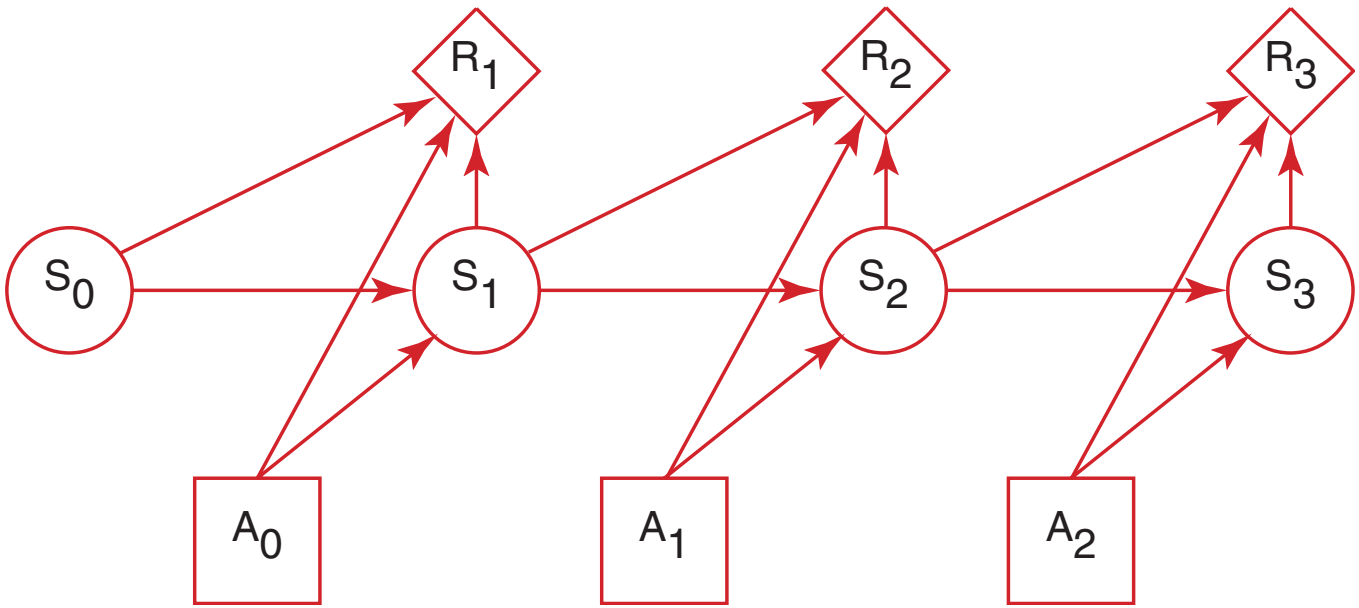
- **Actions**: up, down, left, right.
- **100 states** corresponding to the positions of the robot.
- Robot goes in the commanded direction with probability 0.7, and one of the other directions with probability 0.1.
- If it crashes into an outside wall, it remains in its current position and has a reward of -1.
- Four **special rewarding states**; the agent gets the reward when leaving the state.

Planning Horizons

The planning horizon is how far ahead the planner looks to make a decision.

- The robot gets flung to one of the corners at random after leaving a positive ($+10$ or $+3$) reward state.
  - the process never halts
  - **infinite horizon**
- The robot gets $+10$ or $+3$ entering the state, then it stays there getting no reward. These are **absorbing states.**
  - The robot will eventually reach the absorbing state.
  - **indefinite horizon**

- A **Markov decision process** augments a Markov chain with actions and values (information arcs not shown).



Markov Decision Processes

For an MDP you specify:

- set S of **states**.
- set A of **actions**.
- $P(S_{t+1} \mid S_t, A_t)$ specifies the **dynamics**.
- $R(S_t, A_t, S_{t+1})$ specifies the **reward**. The agent gets a reward at each time step (rather than just a final reward). $R(s, a, s')$ is the expected reward received when the agent is in state s, does action a and ends up in state s'.

Information Availability

What information is available when the agent decides what to do?

- **fully-observable MDP**: the agent gets to observe S_t when deciding on action A_t.
- **partially-observable MDP (POMDP)**: the agent has some noisy sensor of the state.
  It needs to remember its sensing and acting history.
  It can do this by maintaining a sufficiently complex **belief state**.

Rewards and Values

Suppose the agent receives the sequence of rewards $r_1, r_2, r_3, r_4, \ldots$. What value should be assigned?

- **total reward** $V = \sum_{i=1}^{\infty} r_i$
- **average reward** $V = \lim_{n \to \infty} (r_1 + \cdots + r_n)/n$
- **discounted reward** $V = r_1 + \gamma r_2 + \gamma^2 r_3 + \gamma^3 r_4 + \cdots$
  γ is the **discount factor** $0 \leq \gamma \leq 1$.

Policies

- A **stationary policy** is a function:

$$\pi : S \to A$$

  Given a state s, $\pi(s)$ specifies what action the agent who is following π will do.
- An **optimal policy** is one with maximum expected discounted reward.
- For a fully-observable MDP with stationary dynamics and rewards with infinite or indefinite horizon, there is **always an optimal stationary policy.**

Value of a Policy

- $Q^\pi(s, a)$, where a is an action and s is a state, is the expected value of doing a in state s, then following policy π.
- $V^\pi(s)$, where s is a state, is the expected value of following policy π in state s.
- Q^π and V^π can be defined **mutually recursively:**

$$\begin{aligned} Q^\pi(s, a) &= \sum_{s'} P(s' \mid a, s) \left(r(s, a, s') + \gamma V^\pi(s') \right) \\ V^\pi(s) &= Q^\pi(s, \pi(s)) \end{aligned}$$

Value of the Optimal Policy

- $Q^*(s, a)$, where a is an action and s is a state, is the expected value of doing a in state s, then following the **optimal policy**.
- $\pi^*(s)$ is the **optimal action** to take in state s
- $V^*(s)$, where s is a state, is the **expected value of following the optimal policy** in state s.
- Q^* and V^* can be defined mutually recursively:

$$\begin{aligned} Q^*(s, a) &= \sum_{s'} P(s' \mid a, s) \left(r(s, a, s') + \gamma V^*(s') \right) \\ V^*(s) &= \max_a Q^*(s, a) \\ \pi^*(s) &= argmax_a Q^*(s, a) \end{aligned}$$

Value Iteration

- The **t-step lookahead value function**, V^t is the expected value with t steps to go
- Idea: Given an estimate of the t-step lookahead value function, determine the $t+1$-step lookahead value function.

Value Iteration

- Set V^0 arbitrarily, $t = 1$
- Compute Q^t, V^t from V^{t-1}.

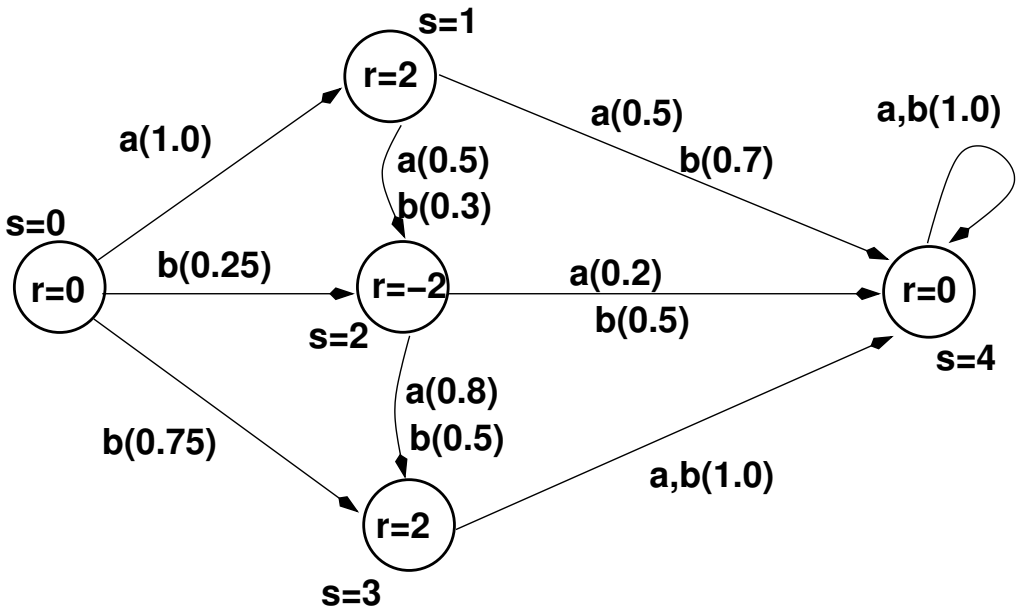
$$Q^t(s,a) = \left[R(s) + \gamma \sum_{s'} Pr(s' \mid s,a) V^{t-1}(s') \right]$$

$$V^t(s) = \max_a Q^t(s,a)$$

- The **policy with t stages** to go is simply the action that maximizes this $\pi^t(s) = \arg\max_a \left[R(s) + \gamma \sum_{s'} Pr(s' \mid s,a) V^{t-1}(s') \right]$
- This is **dynamic programming**
- This **converges** exponentially fast (in t) to the optimal value function.
- **Convergence** when $||V^t(s) - V^{t-1}(s)|| < \epsilon \frac{(1-\gamma)}{\gamma}$ ensures V^t is within ϵ of optimal ($||X|| = max\{|x|, x \in X\}$)

Value Iteration: Simple Example

State space graph (**NOT a DBN**):



Value Iteration: Simple Example

This same graph, represented as a decision network, would have the following factors, where the $(row, col) = (i, j)$ entry in each probability table is $P(S' = j \mid S = i, A)$

$$P(S' \mid S, A = a) = \begin{bmatrix} 0.0 & 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.5 & 0.0 & 0.5 \\ 0.0 & 0.0 & 0.0 & 0.8 & 0.2 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

$$P(S' \mid S, A = b) = \begin{bmatrix} 0.0 & 0.0 & 0.25 & 0.75 & 0.0 \\ 0.0 & 0.0 & 0.3 & 0.0 & 0.7 \\ 0.0 & 0.0 & 0.0 & 0.5 & 0.5 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix} \quad R(S) = \begin{bmatrix} 0 \\ 2 \\ -2 \\ 2 \\ 0 \end{bmatrix}$$

Value Iteration: Simple Example

First iteration, using $\gamma = 0.9$

$$
\begin{aligned}
V^0(s') &= R(s') \\
Q^1(s,a) &= R(s) + \gamma \sum_{s'} P(s' \mid s,a) V^0(s') \\
&= \begin{bmatrix} 1.8 & 1.1 & -0.56 & 2.0 & 0 \\ 0.9 & 1.46 & -1.1 & 2.0 & 0 \end{bmatrix} \\
V^1(s) &= max_a(Q^1(s,a)) \\
&= \begin{bmatrix} 1.8 & 1.46 & -0.56 & 2.0 & 0 \end{bmatrix} \\
\pi^1(s) &= \begin{bmatrix} a & b & a & a & a \end{bmatrix}
\end{aligned}
$$

Value Iteration: Simple Example

Second iteration

$$
\begin{aligned}
Q^2(s,a) &= R(s) + \gamma \sum_{s'} P(s' \mid s,a) V^1(s') \\
&= \begin{bmatrix} 1.31 & 1.75 & -0.56 & 2.0 & 0 \\ 1.22 & 1.85 & -1.1 & 2.0 & 0 \end{bmatrix} \\
V^2(s) &= max_a(Q^2(s,a)) \\
&= \begin{bmatrix} 1.31 & 1.84 & -0.56 & 2.0 & 0 \end{bmatrix} \\
\pi^2(s) &= \begin{bmatrix} a & b & a & a & a \end{bmatrix}
\end{aligned}
$$

on convergence, **optimal value function** is

$$V^*(s) = \begin{bmatrix} 1.66 & 1.85 & -0.56 & 2.0 & 0 \end{bmatrix}$$

policy is

$$\pi^*(s) = \begin{bmatrix} a & b & a & a & a \end{bmatrix}$$

Asynchronous Value Iteration

- You don't need to sweep through all the states, but can update the value function **for each state individually**.
- This converges to the optimal value function, if each state and action is visited **infinitely often** in the limit.
- You can either store $V[s]$ or $Q[s, a]$.

Asynchronous VI: storing $V[s]$

- Repeat forever:
  - Select state s;
  - $V[s] \leftarrow \max_a \sum_{s'} P(s' \mid s, a) \left(R(s, a, s') + \gamma V[s']\right)$;
  - select action a; (e.g. using an exploration policy)

Asynchronous VI: storing $Q[s, a]$

- Repeat forever:
  - Select state s, action a;
  - $Q[s, a] \leftarrow \sum_{s'} P(s' \mid s, a) \left(R(s, a, s') + \gamma \max_{a'} Q[s', a'] \right)$;

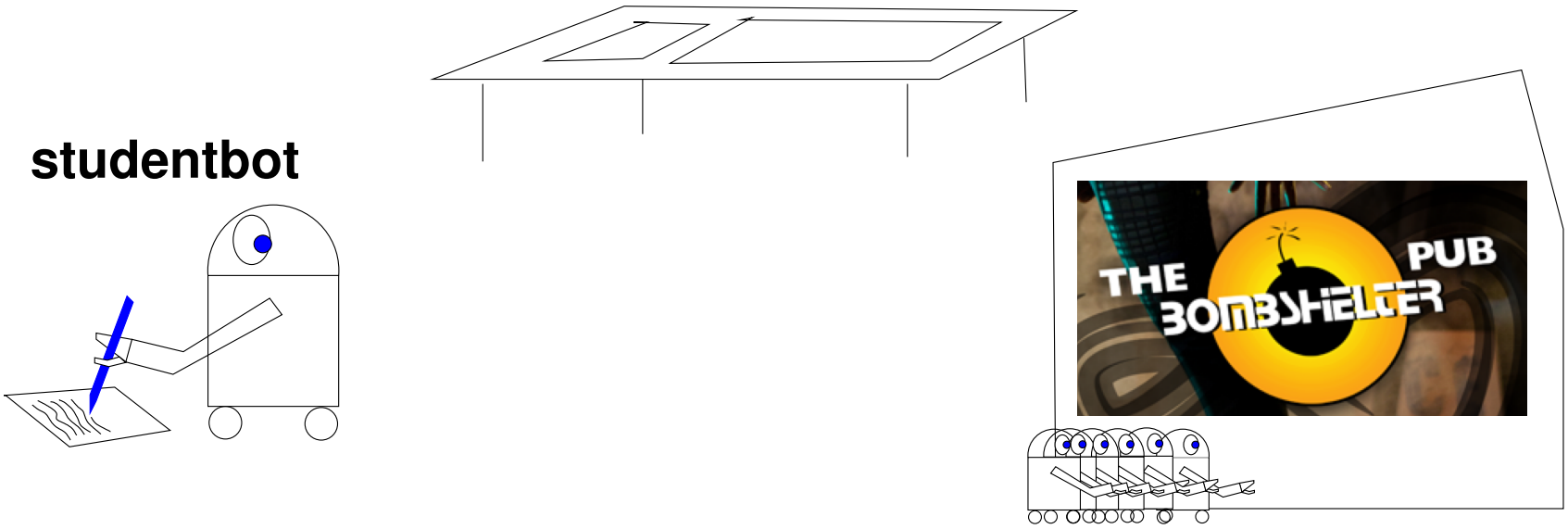
Markov Decision Processes: Factored State

- Represent $S = \{X_1, X_2, \ldots, X_n\}$
- X_i are **random variables**
- for each X_i, and each action $a \in A$, we have $P(X_i' \mid S, A)$
- Reward $R(X_1, X_2, \ldots, X_N)$ may be **additive**:

$$R(X_1, X_2, \ldots, X_N) = \sum_i R(X_i)$$

- Value iteration proceeds as usual but can do one variable at a time (e.g. **variable elimination**)

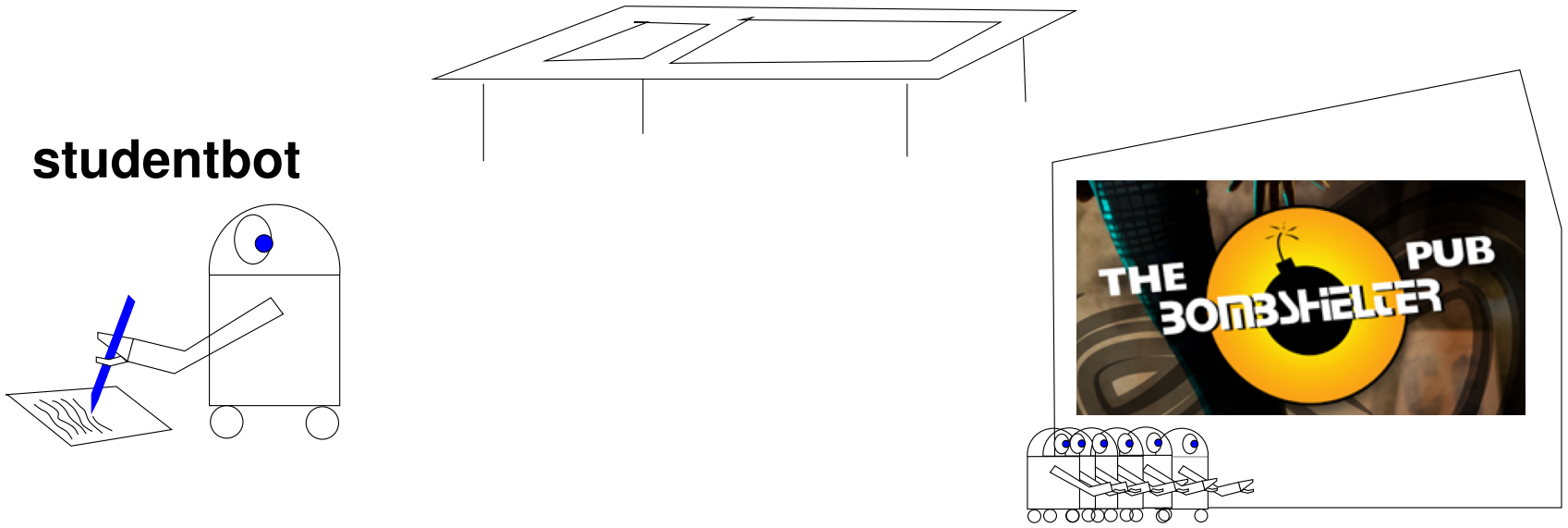
Example: studentbot



state variables (3x2x4x2=48 states):

- **tired**: studentbot is tired (no/a bit/very)
- **passtest**: studentbot passes test (no/yes)
- **knows**: studentbot's state of knowledge (nothing/a bit/a lot/everything)
- **goodtime**: studentbot has a good time (no/yes)

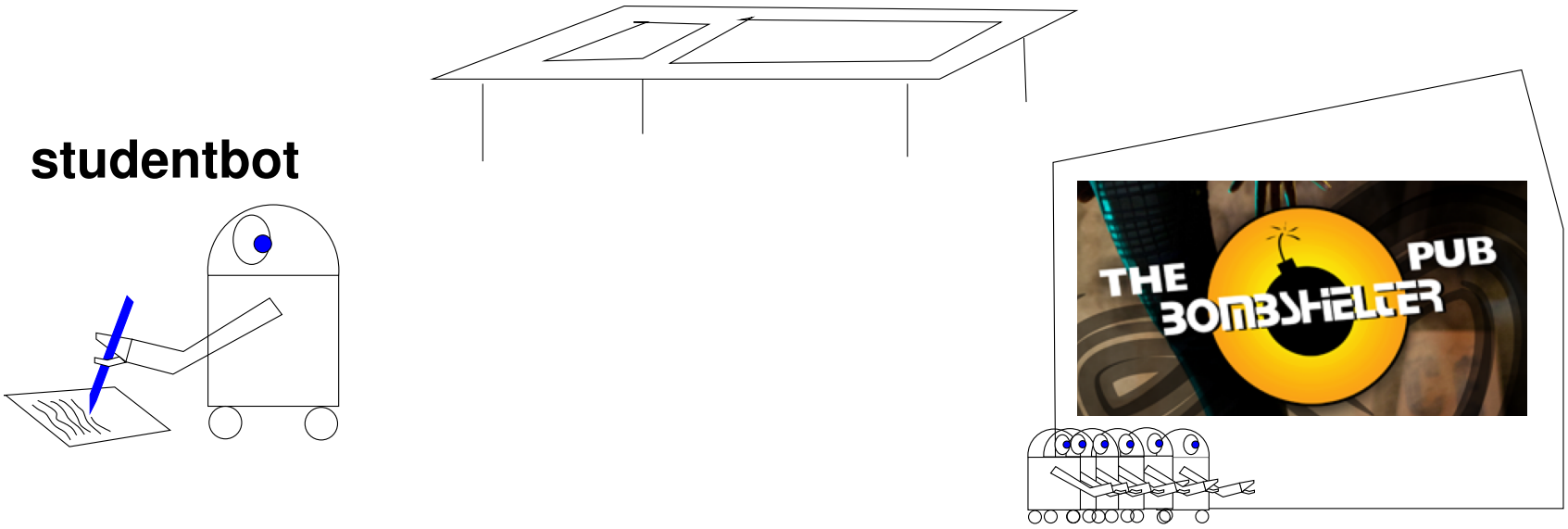
Example: studentbot



studentbot actions:

- **study**: studentbot's knowledge increases, studentbot gets tired
- **sleep**: studentbot gets less tired
- **party**: studentbot has a good time if not tired, but gets tired and loses knowledge
- **take test**: studentbot takes a test (can take test anytime)

Example: studentbot



studentbot rewards:

- +20 if studentbot passes the test
- +2 if studentbot has a good time

basic tradeoff: short term vs. long-term rewards

Studentbot

State-based:

$$P(s' \mid s, a) = [48 \times 48]$$

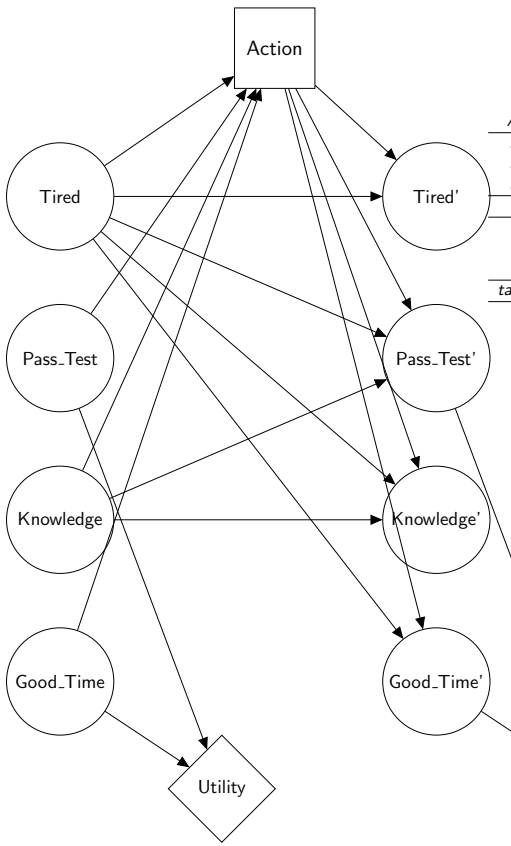
$$R(s) = [48 \times 1]$$

Studentbot

As a **dynamic decision network**:

Action	Tired	Knowledge	Knowledge′ nothing	a_bit	a_lot	everything
study	no	nothing	0.5	0.5	0	0
	no	a_bit	0	0.5	0.5	0
	no	a_lot	0.0	0.0	0.5	0.5
	no	everything	0		0	1.0
	a_bit	nothing	0.5	0.5	0	0
	a_bit	a_bit	0	0.5	0.5	0.0
	a_bit	a_lot	0	0	0.5	0.5
	a_bit	everything	0	0	0	1.0
	very	nothing	1.0	0	0	0
	very	a_bit	0	1.0	0.0	0.0
	very	a_lot	0	0	1.0	0.0
	very	everything	0	0	0	1.0
party	–	nothing	1.0	0.0	0	0
	no	a_bit	0.5	0.5	0.0	0.0
	no	a_lot	0.0	0.5	0.5	0.0
	no	everything	0	0	0.5	0.5
sleep	–	nothing	1.0	0.0	0	0
	–	a_bit	0.0	1.0	0.0	0.0
	–	a_lot	0.0	0.0	1.0	0.0
	–	everything	0	0.0	0.0	1.0
take_test	–	–	1.0	0.0	0	0

Action	Tired	Good_Time′ no	yes
party	no	0.0	1.0
party	a_bit / very	1.0	0.0
other	–	1.0	0.0



Action	Tired	Tired′ no	a_bit	very
study	no	0.8	0.2	0.0
study	abit	0.0	0.8	0.2
study	very	0.0	0.0	1.0
sleep	–	1	0	0
party	no	0	1	0
party	a_bit	0	0	1
party	a_lot	0	0	1
take_test	–	0	0	1

Action	Tired	Knowledge	PassTest′ no	yes
take_test	no	nothing	1	0
	no	a_bit	0.9	0.1
	no	a_lot	0.7	0.3
	no	everything	0.1	0.9
	a_bit	nothing	1	0
	a_bit	a_bit	0.9	0.1
	a_bit	a_lot	0.7	0.3
	a_bit	everything	0.1	0.9
	very	–	1	0
else	–	–	1	0

Good_Time	Pass_Test	U(Good_Time, Pass_Test)
yes	yes	22
yes	no	2
no	yes	20
no	no	0

StudentbotValues and Policy

tired	know	policy	value
no	nothing	party	2
no	a bit	party	2
no	a lot	take test	6
no	everything	take test	18
a bit	nothing	study	0
a bit	a bit	take test	2
a bit	a lot	take test	6
a bit	everything	take test	8
very	nothing	study	0
very	a bit	study	0
very	a lot	study	0
very	everything	study	0

After one iteration

policy	value
study	16.7
study	20.8
study	25.7
take test	31.6
study	15.97
study	19.94
study	26.0
take test	31.6
sleep	15.1
sleep	18.7
sleep	23.1
sleep	28.4

Optimal to within 0.01 (72 iterations)

Partially Observable Markov Decision Processes (POMDPs)

A **POMDP** is like an MDP, but

some variables are **not observed**. It is a tuple $\langle S, A, T, R, O, \Omega \rangle$

More details in handout `mdp.pdf`

S: finite set of **unobservable states**

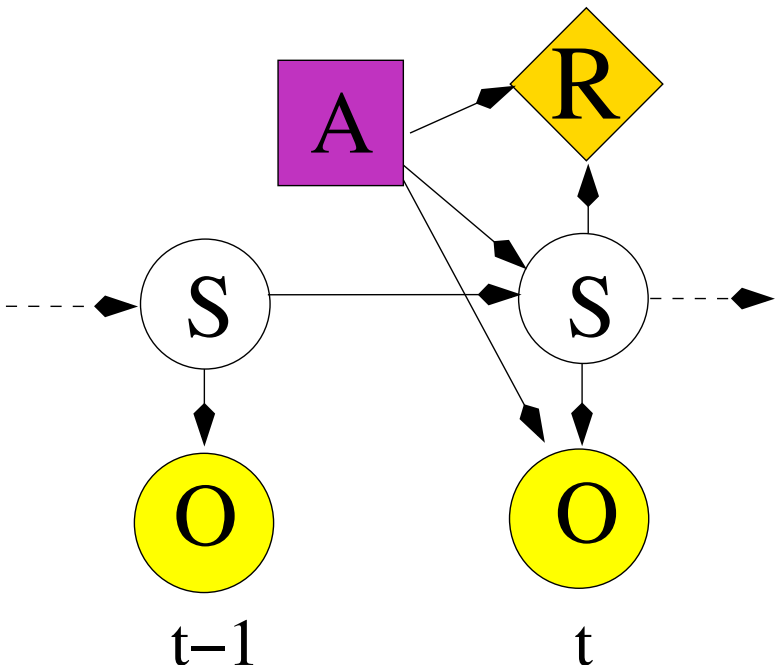
A: finite set of agent **actions**

$T : S \times A \to S$ **transition function**

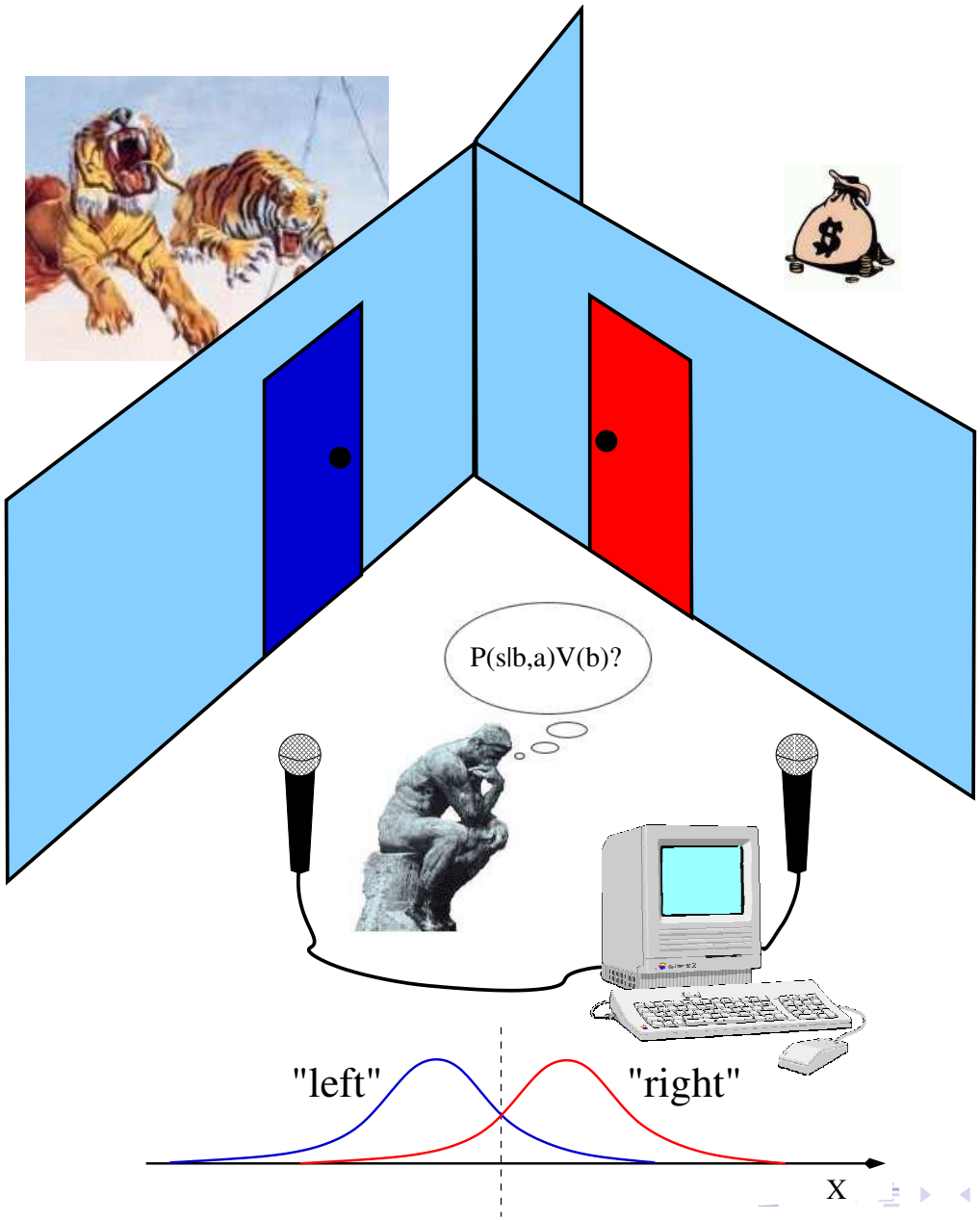
$R : S \times A \to \mathcal{R}$ **reward** function

O: set of **observations**

$\Omega : S \times A \to O$ **observation function**



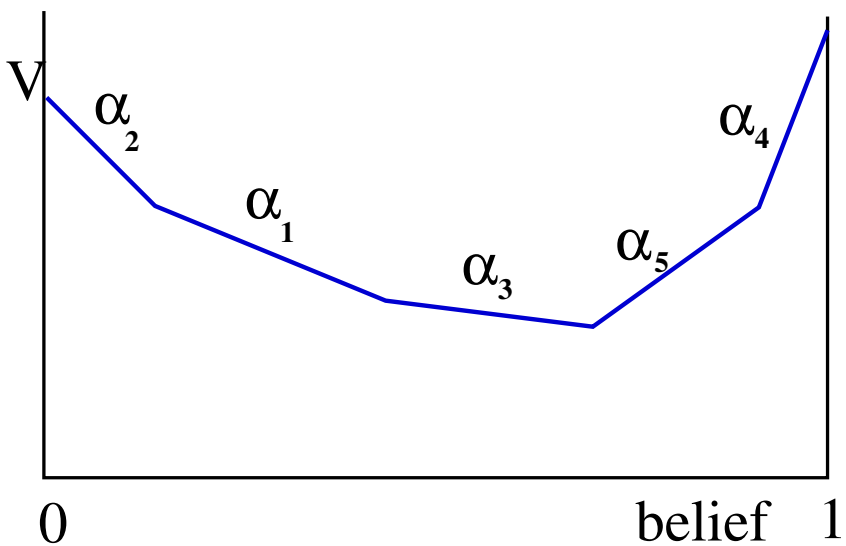
e.g. 1-D Tiger problem

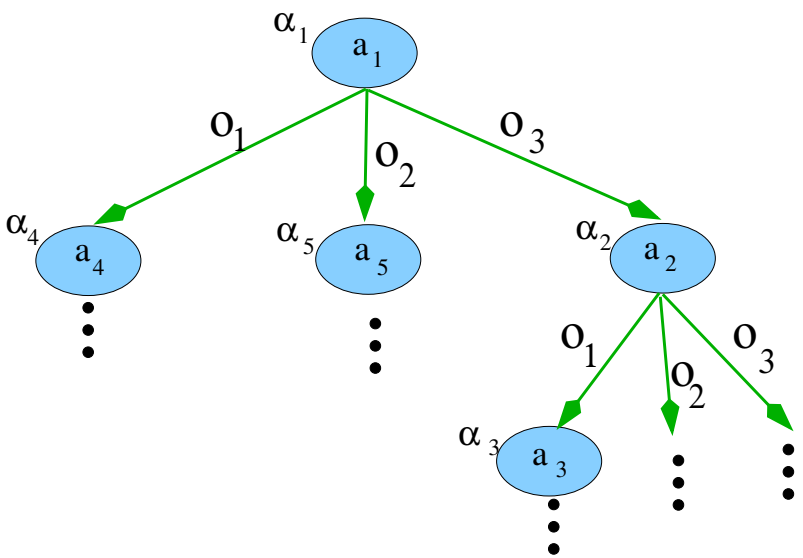


Value Functions and Conditional Plans

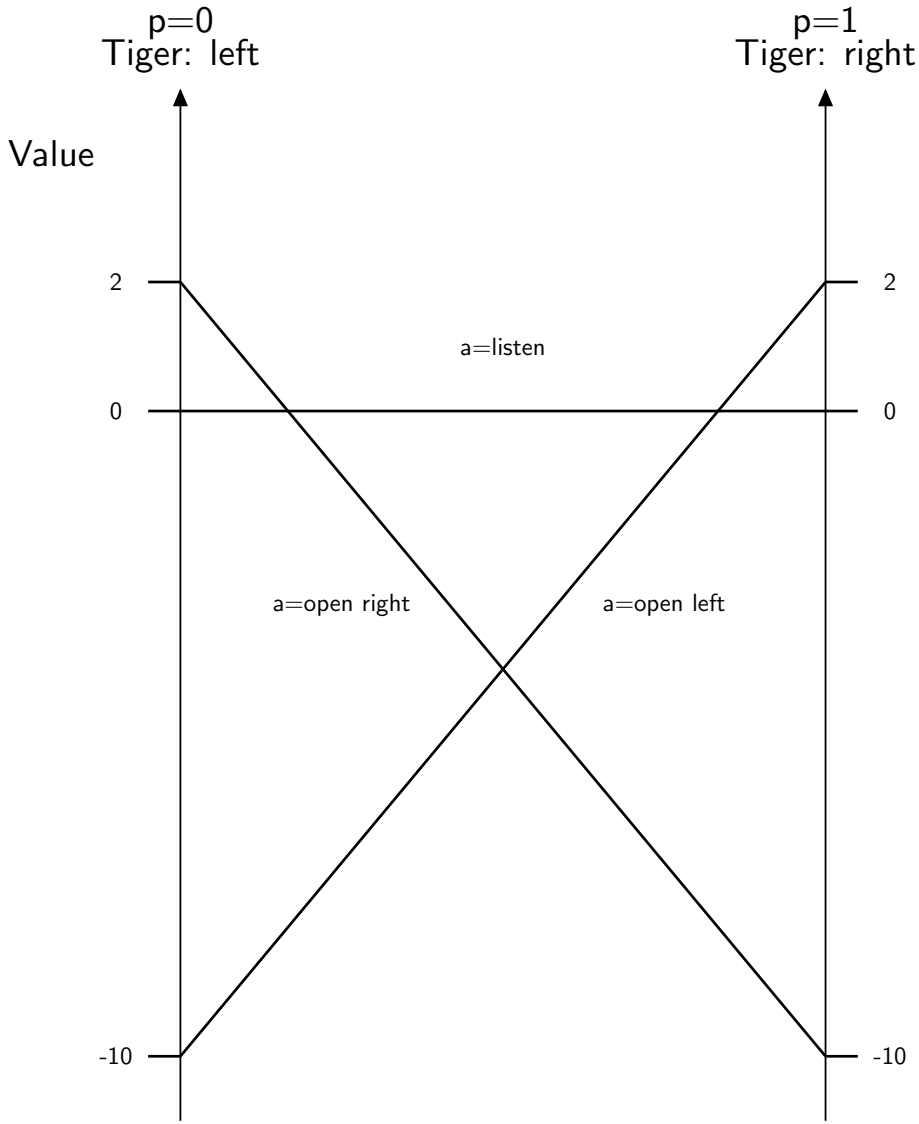
$$V^{k+1}(b) = \max_a R^a(b) + \gamma \sum_o Pr(o|b,a) V^k(b_o^a)$$

$V(b)$ can be represented with a piecewise linear function over the belief space - pieces are called α vectors

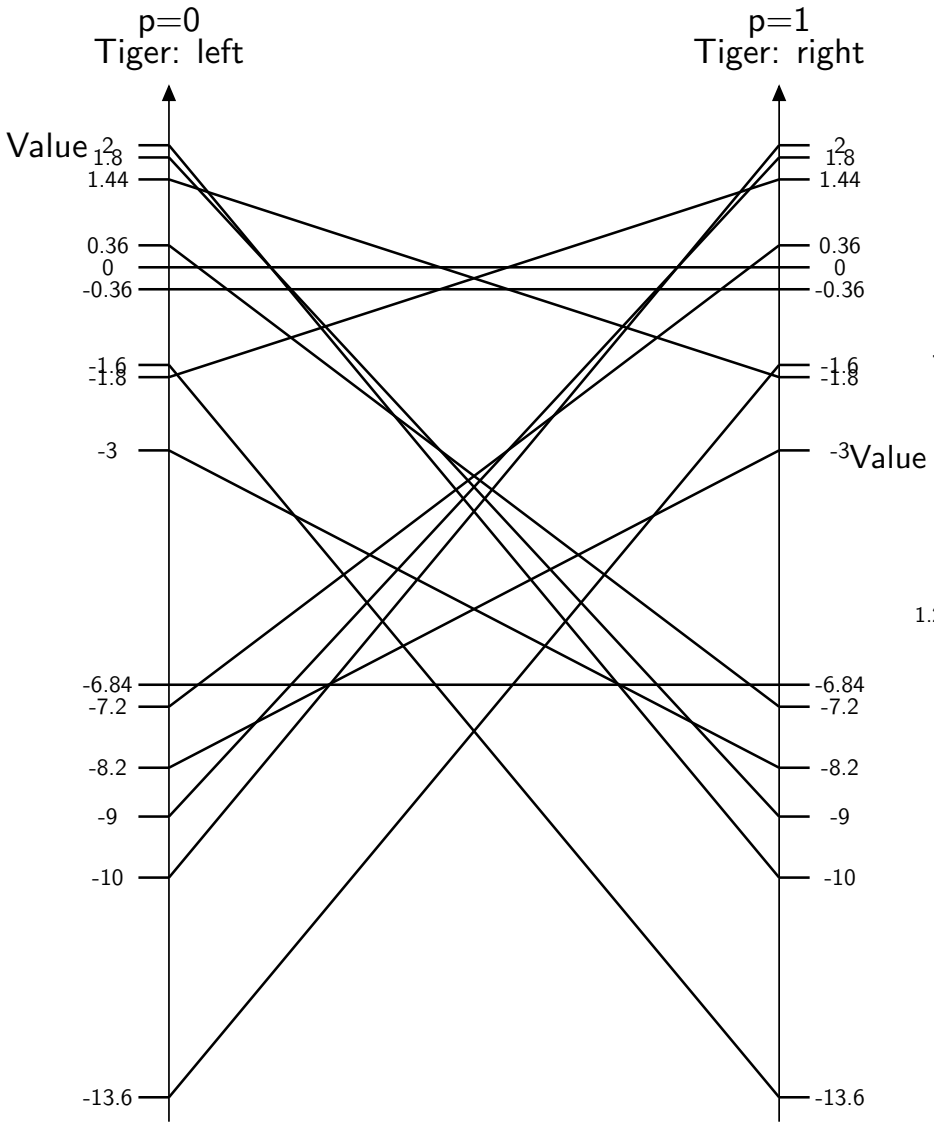




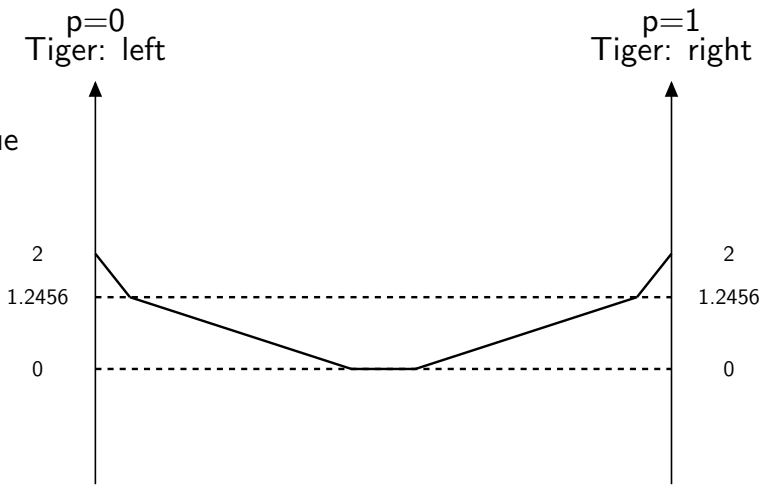
e.g. Tiger problem, after zero iterations



e.g. Tiger problem, after one iteration



all generated α vectors



optimal value function

Policies

Policy: maps beliefs states into actions $\pi(b(s)) \rightarrow a$

Two ways to compute a policy

1. **Backwards search**
   - **Dynamic programming** (Variable Elimination)
   - in MDP: $Q_t(s,a) = R(s,a) + \gamma \sum_{s'} Pr(s'|s,a) \max_{a'} Q_{t-1}(s',a')$
   - in POMDP: $Q_t(b(s),a)$
   - **Point-based** backups make this efficient
2. **Forwards search**
   - Expand beliefs going forward
   - $b_o^a(s')$ is reached (from $b(s)$) after taking action a and observing o using Transition T and Observation Ω functions:

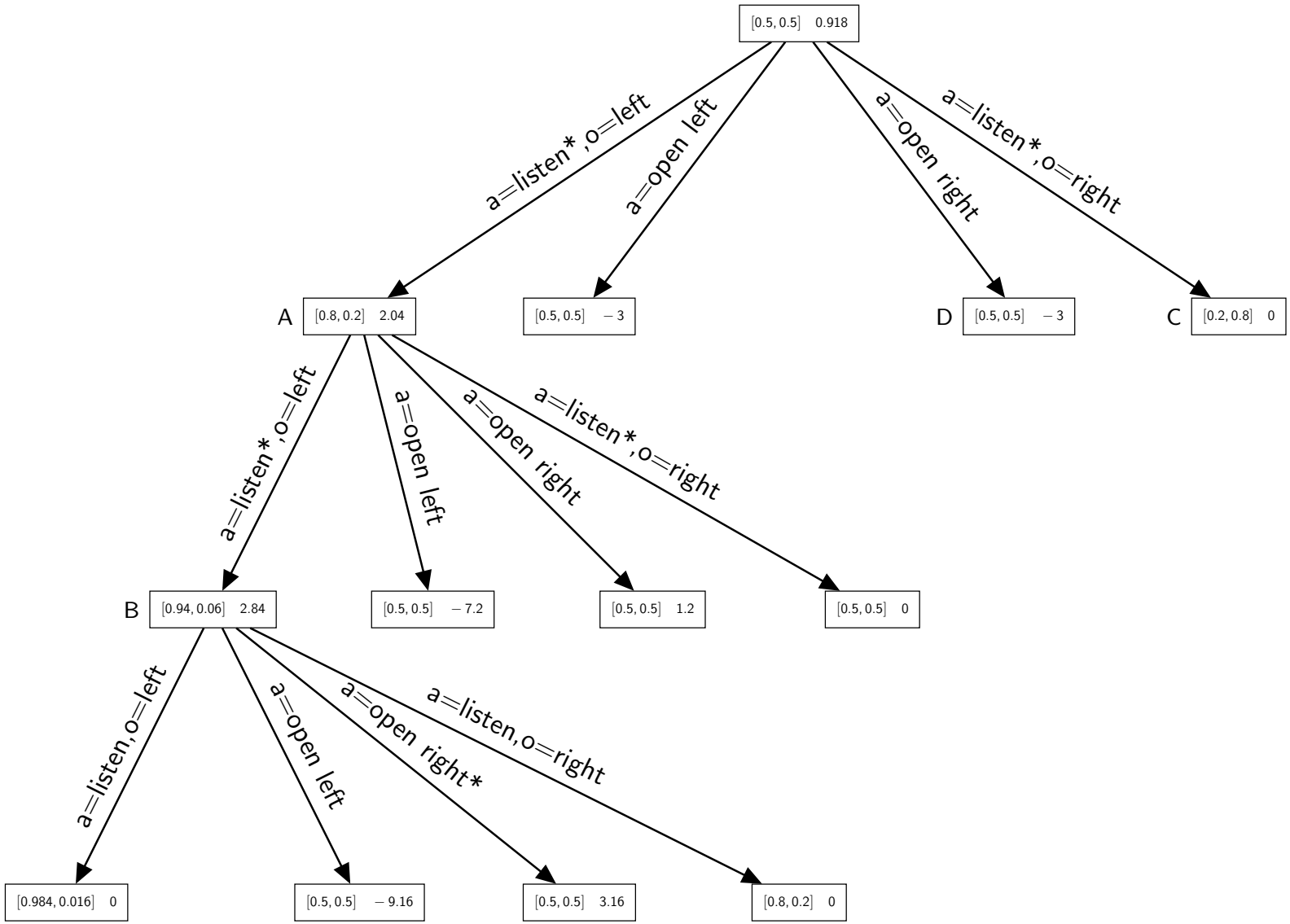
$$b_o^a(s') = \sum_{s \in S} T(s'|a,s)\Omega(o|s',a)b(s) \quad \forall b \in \mathcal{B}$$

   - Can expand more deeply in **promising directions**, and
   - Ensure **exploration** using Monte-Carlo Tree Search (**MCTS**)

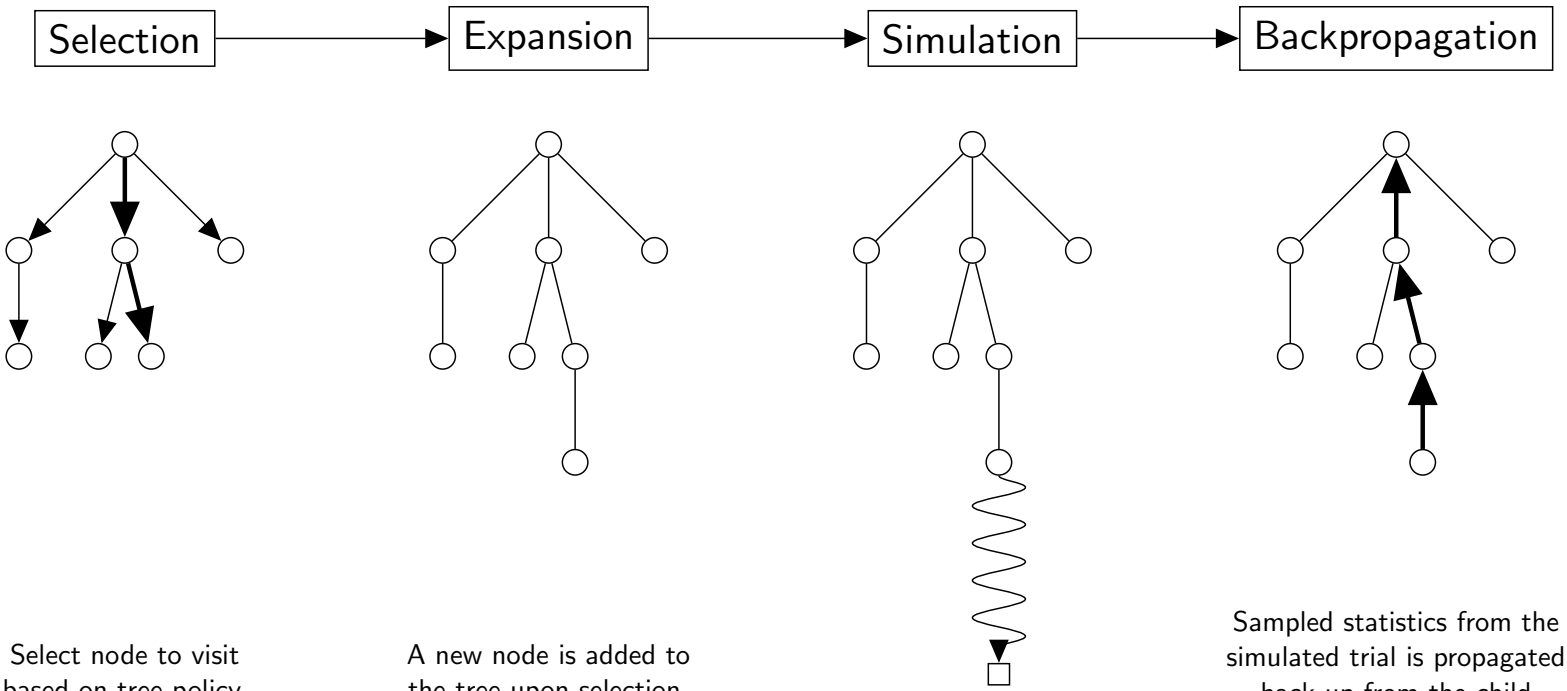
Forward Search for POMDPs

procedure GetValue($b(s)$)
for each action-observation pair a, o:
    $b_o^a(s') \leftarrow$ propagate $b(s)$ forwards for each action and
        observation using stochastic simulation/particle filter
    if $b_o^a(s')$ not at a leaf:
        evaluate recursively by further growing the tree:
            $V_o^a \leftarrow GetValue(b_o^a(s'))$
    else:
        create a new leaf for a, o
            do a series of single-belief point rollouts
            (e.g. propagate a single belief forward stochastically
            gathering reward until termination condition is met),
            use the total returned value as V_o^a.
return $R(b(s)) + \max_a \{\gamma \sum_o P(o|b(s), a) \sum_{s'} V_o^a b_o^a(s'))\}$

e.g. Tiger problem, two steps expanded



MCTS



Select node to visit based on tree policy.

A new node is added to the tree upon selection.

Run trial simulation based on a default policy (usually random) from the newly created node until terminal node is reached.

Sampled statistics from the simulated trial is propagated back up from the child nodes to the ancestor nodes.

Next

- Reinforcement Learning (Poole & Mackworth chapter 13.1-12.9)